KAESER KOMPRESSOREN

Technical Description
SIGMA CONTROL 2

Process Map ≥ 6.4.1

≥ fluid_6.4.1 & ≥ vac_6.4.1

7_7601_PA_27E

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General

1. In communication mode "send/receive" Sigma Control can monitor the master program (factory setting). This takes place with "iSlaveMsgHeader" and "iSlaveMsgFooter" that have to be returned unchanged from master.
   Sigma Control ignores the remote control signals (Table 2) if these values do not return from the master within the monitoring time (timeout, factory setting 5s). A message is shown on the display. As soon as the values arrive correctly again, the remote control signals are taken into account again.

   Example sequence for Profibus Master in Step 7 :

   L DB (receive box master).DBW 94 Message header
   T DB (transmit box master).DBW 38

   L DB (receive box master).DBW 126 Message footer
   T DB (transmit box master).DBW 62

   Montioring can be deactivated in menu *"Communication / Com-Module / …… / Communication error: Timeout"* by removing the check mark at check box right of Timeout.
   If monitoring is not required the above sequence is not used.

2. Availability or use of some signals depends on type of compressor, compressor options or use of functions at SC2. Information see column "Availability" or "Usability" in the following tables in combination with table 3.
   For further information on signal availability and use ask your local KAESER subsidary or KAESER Headquarter Service Support Engineering, phone +49 9561 640-7979.

Sigma Control 2 Profibus

1. The data required for planning a Profibus project are in the equipment specification file "KAES0CEC.gsd".

2. The Sigma Control is a Profibus DPV0 slave with 64 bytes of input data and 128 bytes of output data. Depending on the number of Sigma Control units used the Profibus DP master must have a correspondingly large enough address space.

Sigma Control 2 Profinet

1. The data required for planning a Profinet project are in the equipment specification file "GSDML-V2.25-Kaeser-SC2-20120203.xml".

Sigma Control 2 Modbus

1. **Compressor ==>>>> Master Controller**
   Read Input Registers with modbus function 04 "Read Input Registers" or read Holding Registers with modbus function 03 "Read Holding Registers".
   Modbus Protocol address: Address of the data at the bus (Base 0).
   Modbus Register: Register address of the data = PLC-address (Base 1).
   3xxxx **3** = Identification of Input Register, xxxx = Register address
   4xxxx **4** = Identification of Holding Register, xxxx = Register address

2. **Master Controller ==>>>> Compressor**
   Write registers with function 16 "Write Multiple Registers" or 06 "Write Single Register".
   Read registers with function 03 "Read Holding Registers". Never written registers contain "0".

Modbus Protocol address:	Address of the data at the bus (Base 0).
Modbus Register:	Register address of the data = PLC-address (Base 1).
4xxxx	**4** = Identification of Holding Register, xxxx = Register address

Sigma Control 2 Devicenet

1. The data required for planning a DeviceNet project are in the equipment specification file "324-8172-EDS_ABCC_DEV_V_2_3_1.eds".
2. 8 Bit register: Low bytes are shown at low address.

Sigma Control 2 EtherNet/IP

1. The data required for planning an EtherNet/IP project are in the equipment specification file "SC2_EtherNetIP.eds".
2. 8 Bit register: Low bytes are shown at low address.

SAM Send/Receive

1. Data for compressor 1 to 16 can be found in SAM data block 64 to 79.

1. Compressor ==>>>> Master Controller

Availability	
4	see associated table (1.x)
3	depends on use of the function at SC 2
2.x	see table 3
1	always

Table 1: Process data

* -32768 = Value invalid or not used

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

Sigma Control 2								SAM		Description	Avalability	Unit	min	max	Format
JSON-RP	PROFIBUS/ PROFINET		Modbus**				DeviceNet EtherNet/IP	Send/Receiv							
			Input		Holding										
BDSnd	S7 DBW	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	Register MSB/LSB	S5 DW	S7 DBW						
	0	0	21	30022	277	40278	1/0	32	64	Alarm xAM16 - xAM1 Table 1.1	4	binary			BOOL
	2	1	22	30023	278	40279	3/2	33	66	Alarm xAM32 - xAM17 Table 1.1	4	binary			BOOL
	4	2	23	30024	279	40280	5/4	34	68	Alarm xAM48 - xAM33 Table 1.1	4	binary			BOOL
	6	3	24	30025	280	40281	7/6	35	70	Alarm xAM64 - xAM49 Table 1.1	4	binary			BOOL
	8	4	25	30026	281	40282	9/8	36	72	Alarm xAM80 - xAM65 Table 1.1	4	binary			BOOL
	10	5	26	30027	282	40283	11/10	37	74	Warning xWM16 - xWM1 Table 1.2	4	binary			BOOL
	12	6	27	30028	283	40284	13/12	38	76	Warning xWM32 - xWM17 Table 1.2	4	binary			BOOL
	14	7	28	30029	284	40285	15/14	39	78	Warning xWM48 - xWM33 Table 1.2	4	binary			BOOL
	16	8	29	30030	285	40286	17/16	40	80	Warning xWM64 - xWM49 Table 1.2	4	binary			BOOL
	18	9	30	30031	286	40287	19/18	41	82	Warning xWM80 - xWM65 Table 1.2	4	binary			BOOL
	20	10	31	30032	287	40288	21/20	42	84	Operational message xOM16 - xOM1 Table 1.3	4	binary			BOOL
	22	11	32	30033	288	40289	23/22	43	86	Operational message xOM32 - xOM17 Table 1.3	4	binary			BOOL
	24	12	33	30034	289	40290	25/24	44	88	Operational message xOM40 - xOM33 Table 1.3	4	binary			BOOL
	26	13	34	30035	290	40291	27/26	45	90	Binary signal xBS15 - xBS0 Table 1.4	4	binary			BOOL
	28	14	35	30036	291	40292	29/28	46	92	Binary signal xBS31 - xBS16 Table 1.4	4	binary			BOOL
	30	15	36	30037	292	40293	31/30	47	94	Binary signal xBS47 - xBS32 Table 1.4	4	binary			BOOL
	32	16	37	30038	293	40294	33/32	48	96	Binary signal xBS63 - xBS48 Table 1.4	4	binary			BOOL
iDV0	34	17	38	30039	294	40295	35/34	49	98	Software version 3rd digit / 4th digit	1				BYTE/BYTE
iDV1	36	18	39	30040	295	40296	37/36	50	100	Software version 1st digit / 2nd digit	1				BYTE/BYTE
iDV2	38	19	40	30041	296	40297	39/38	51	102	Remaining idle time of the idle warming	3	s	*		INT
iDV3	40	20	41	30042	297	40298	41/40	52	104	Pressure actual value pNloc	1	mbar	-400 *	20000	INT
iDV4	42	21	42	30043	298	40299	43/42	53	106	Internal pressure pi	2.1	mbar	-400 *	20000	INT
iDV5	44	22	43	30044	299	40300	45/44	54	108	Airend discharge temperature ADT	1	0.1 °C	-500 *	2999	INT
iDV6	46	23	44	30045	300	40301	47/46	55	110	Refrigeration dryer air temperature	2.T	0.1 °C	-500 *	2999	INT

1. Compressor ==>>>> Master Controller

4	see associated table (1.x)
3	depends on use of the function at SC 2
2.x	see table 3
1	always

Table 1: Process data

* -32768 = Value invalid or not used

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

Sigma Control 2									SAM							
JSON-RP	PROFIBUS/ PROFINET		Modbus**				DeviceNet EtherNet/IP		Send/Receiv		Description	Availability	Unit	min	max	Format
			Input		Holding											
BDSnd	S7 DBW	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	Register MSB/LSB		S5 DW	S7 DBW						
iDV7	48	24	45	30046	301	40302	49/48		56	112	Inlet temperature	2.2	0.1 °C	-500 *	2999	INT
iDV8	50	25	46	30047	302	40303	51/50		57	114	Pressure discharge temperature	2.3	0.1 °C	-500 *	2999	INT
iDV9	52	26	47	30048	303	40304	53/52		58	116	Oil separator air temperature	2.4	0.1 °C	-500 *	2999	INT
iDV10	54	27	48	30049	304	40305	55/54		59	118	Motor temperature	2.5	0.1 °C	-500 *	2999	INT
iDV11	56	28	49	30050	305	40306	57/56		60	120	Allocatable with AOI2.00 or p, T, I at menu 5.7.2.4.x	2.24		*		INT
iDV12	58	29	50	30051	306	40307	59/58		61	122	Oil separator differential pressure	2.1	mbar	-400 *	20000	INT
iDV13	60	30	51	30052	307	40308	61/60		62	124	Pressure actual value pN (pressure regulation)	1	mbar	-400 *	20000	INT
iDV14	62	31	52	30053	308	40309	63/62		63	126	Current drive motor starts/h	1	1			INT
iDV15	64	32	53	30054	309	40310	65/64		64	128	Total compressor hours run x 1000	1	h			INT
iDV16	66	33	54	30055	310	40311	67/66		65	130	Total compressor hours run x 1	1	h			INT
iDV17	68	34	55	30056	311	40312	69/68		66	132	Hours on load x 1000	1	h			INT
iDV18	70	35	56	30057	312	40313	71/70		67	134	Hours on load x 1	1	h			INT
iDV19	72	36	57	30058	313	40314	73/72		68	136	Setpoint pressure pA switching point	1	mbar			INT
iDV20	74	37	58	30059	314	40315	75/74		69	138	Setpoint pressure pA switching difference	1	mbar			INT
iDV21	76	38	59	30060	315	40316	77/76		70	140	Setpoint pressure pB switching point	1	mbar			INT
iDV22	78	39	60	30061	316	40317	79/78		71	142	Setpoint pressure pB switching difference	1	mbar			INT
iDV23	80	40	61	30062	317	40318	81/80		72	144	Nominal package pressure	1	mbar			INT
iDV24	82	41	62	30063	318	40319	83/82		73	146	Increased cut-out pressure pE	1	mbar			INT
iDV25	84	42	63	30064	319	40320	85/84		74	148	Remaining drive motor idle time	1	s			INT
iDV26	86	43	64	30065	320	40321	87/86		75	150	Allocatable with p, T, I at SC 2 menu 5.7.2.4.x	3		*		INT
iDV27	88	44	65	30066	321	40322	89/88		76	152	Allocatable with p, T, I at SC 2 menu 5.7.2.4.x	3		*		INT
iDV28	90	45	66	30067	322	40323	91/90		77	154	Allocatable at menu 3.3 (kWh x 1000) or at 5.7.2.4.x	2.C3		*		INT
iDV29	92	46	67	30068	323	40324	93/92		78	156	Allocatable at menu 3.3 (kWh x 1) or at 5.7.2.4.x	2.C3		*		INT
iSlaveMsgHeader	94	47	68	30069	324	40325	95/94		79	158	Message header slave	1				WORD

1. Compressor ==>>>> Master Controller

4	see associated table (1.x)
3	depends on use of the function at SC 2
2.x	see table 3
1	always

Table 1: Process data

* -32768 = Value invalid or not used

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

Sigma Control 2								SAM		Description	Availability	Unit	min	max	Format
JSON-RP	PROFIBUS/ PROFINET		Modbus**				DeviceNet EtherNet/IP	Send/Receiv							
			Input		Holding										
BDSnd	S7 DBW	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	Register MSB/LSB	S5 DW	S7 DBW						
iDV30	96	48	69	30070	325	40326	97/96	80	160	Remaining service life oil seperator cartridge	1	h			INT
iDV31	98	49	70	30071	326	40327	99/98	81	162	Remaining service life oil	1	h			INT
iDV32	100	50	71	30072	327	40328	101/100	82	164	Remaining service life oil filter	1	h			INT
iDV33	102	51	72	30073	328	40329	103/102	83	166	Remaining service life air filter	1	h			INT
iDV34	104	52	73	30074	329	40330	105/104	84	168	Remaining time till inspection of the valves	1	h			INT
iDV35	106	53	74	30075	330	40331	107/106	85	170	Remaining time till inspection of V-belts/coupling	1	h			INT
iDV36	108	54	75	30076	331	40332	109/108	86	172	Remaining service life motor bearings	1	h			INT
iDV37	110	55	76	30077	332	40333	111/110	87	174	Remaining time till inspection of electrical equipment	1	h			INT
iDV38	112	56	77	30078	333	40334	113/112	88	176	Remaining time till relubrication of bearings	1	h			INT
iDV39	114	57	78	30079	334	40335	115/114	89	178	Current cut-out pressure pA (pB) switching point	1	mbar			INT
iDV40	116	58	79	30080	335	40336	117/116	90	180	Current cut-out pressure pA (pB) switching difference	1	mbar			INT
iDV41	118	59	80	30081	336	40337	119/118	91	182	Compressor model / free	1				INT
iDV42	120	60	81	30082	337	40338	121/120	92	184	Compressor start delay	3	s			INT
iDV43	122	61	82	30083	338	40339	123/122	93	186	Duty cycle	1	1%			INT
iDV44	124	62	83	30084	339	40340	125/124	94	188	Current motor speed	2.F	1/min			INT
iSlaveMsgFooter	126	63	84	30085	340	40341	127/126	95	190	Message footer slave	1				WORD

1. Compressor ==>>>> Master Controller

M	Modbus only
P	Profibus only
1*	since manuf. date MFGDT: 2010/11
1	always

Tabelle 1: Diagnostic data

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

Sigma Control 2									SAM		Description	Availability	Example		
JSON-RP	PROFIBUS/ PROFINET		Modbus**					DeviceNe EtherNet/IP	Send/Receiv						
			Input		Holding										
BDSnd	S7 DBW	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	Byte	Register MSB/LSB	S5 DW	S7 DBB			HEX	Value	Result
%	0	DL0	%	%	%	%		%	DL96	192	Profibus: System status 1	P			
%	1	DR0	%	%	%	%		%	DR96	193	Profibus: System status 2	P			
%	2	DL1	%	%	%	%		%	DL97	194	Profibus: System status 3	P			
%	3	DR1	%	%	%	%		%	DR97	195	Profibus: DP master address	P	01	1	
%	4	DL2	%	%	%	%		%	DL98	196	Profibus: Manufacturer code: PNO number	P	0C		Kaeser PNO Nr.:
%	5	DR2	%	%	%	%		%	DR98	197	Profibus: Manufacturer code: PNO number	P	EC		**0CEC**
%	6	DL3	%	%	%	%		%	DL99	198	Profibus: Header-Byte specific diagnostic data	P			
%	7	DR3	1	30002	257	40258	MSB	%	DR99	199	SC2 MCS Kaeser SN: 5th digit, 6th digit *	1*	62	62	SC2 MCS Kaeser SN: **003062**
%	8	DL4	1	30002	257	40258	LSB	%	DL100	200	SC2 MCS Kaeser SN: 3rd digit, 4th digit *	1*	30	30	
%	9	DR4	2	30003	258	40259	MSB	%	DR100	201	SC2 MCS Kaeser SN: 1st digit, 2nd digit *	1*	00	00	
%	10	DL5	2	30003	258	40259	LSB	%	DL101	202	reserved		00		
%	11	DR5	8	30009	264	40265	MSB	%	DR101	203	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	SC2 MCS Kaeser PN: **7.7601P0**
%	12	DL6	8	30009	264	40265	LSB	%	DL102	204	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	
%	13	DR6	9	30010	265	40266	MSB	%	DR102	205	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	
%	14	DL7	9	30010	265	40266	LSB	%	DL103	206	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	
%	15	DR7	10	30011	266	40267	MSB	%	DR103	207	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	
%	16	DL8	10	30011	266	40267	LSB	%	DL104	208	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	
%	17	DR8	11	30012	267	40268	MSB	%	DR104	209	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	
%	18	DL9	11	30012	267	40268	LSB	%	DL105	210	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	
%	19	DR9	4	30005	260	40261	MSB	%	DR105	211	SC2 MCS Kaeser PN: 8th digit (ASCII)	1	30	0	
%	20	DL10	4	30005	260	40261	LSB	%	DL106	212	SC2 MCS Kaeser PN: 7th digit (ASCII)	1	50	P	
%	21	DR10	5	30006	261	40262	MSB	%	DR106	213	SC2 MCS Kaeser PN: 6th digit (ASCII)	1	31	1	
%	22	DL11	5	30006	261	40262	LSB	%	DL107	214	SC2 MCS Kaeser PN: 5th digit (ASCII)	1	30	0	
%	23	DR11	6	30007	262	40263	MSB	%	DR107	215	SC2 MCS Kaeser PN: 4th digit (ASCII)	1	36	6	

1. Compressor ==>>>> Master Controller

M	Modbus only
P	Profibus only
1*	since manuf. date MFGDT: 2010/11
1	always

Tabelle 1: Diagnostic data

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

JSON-RP	Sigma Control 2									SAM		Description	Avalability	Example		
	PROFIBUS/ PROFINET		Modbus**					DeviceNe EtherNet/IP	Send/Receiv							
			Input		Holding											
BDSnd	S7 DBW	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	Byte	Register MSB/LSB	S5 DW	S7 DBB			HEX	Value	Result
%	24	DL12	6	30007	262	40263	LSB	%	DL108	216	SC2 MCS Kaeser PN: 3rd digit (ASCII)	1	37	7	
%	25	DR12	7	30008	263	40264	MSB	%	DR108	217	SC2 MCS Kaeser PN: 2nd digit (ASCII)	1	2E	.	
%	26	DL13	7	30008	263	40264	LSB	%	DL109	218	SC2 MCS Kaeser PN: 1st digit (ASCII)	1	37	7	
%	27	DR13	12	30013	268	40269	MSB	%	DR109	219	reserved		00		
%	28	DL14	12	30013	268	40269	LSB	%	DL110	220	reserved		00		
%	29	DR14	13	30014	269	40270	MSB	%	DR110	221	reserved		00		
%	30	DL15	13	30014	269	40270	LSB	%	DL111	222	reserved		00		
%	%	%	14	30015	270	40271	MSB	%	%	%	reserved		00		
%	%	%	14	30015	270	40271	LSB	%	%	%	reserved		00		
%	%	%	15	30016	271	40272	MSB	%	%	%	reserved		00		
%	%	%	15	30016	271	40272	LSB	%	%	%	reserved		00		
%	%	%	16	30017	272	40273	MSB	%	%	%	reserved		00		
%	%	%	16	30017	272	40273	LSB	%	%	%	reserved		00		
%	%	%	17	30018	273	40274	MSB	%	%	%	reserved		00		
%	%	%	17	30018	273	40274	LSB	%	%	%	reserved		00		
%	%	%	18	30019	274	40275	MSB	%	%	%	Modbus: Controller identifier (ASCII)	M	20	SP	SC2
%	%	%	18	30019	274	40275	LSB	%	%	%	Modbus: Controller identifier (ASCII)	M	53	S	
%	%	%	19	30020	275	40276	MSB	%	%	%	Modbus: Controller identifier (ASCII)	M	43	C	
%	%	%	19	30020	275	40276	LSB	%	%	%	Modbus: Controller identifier (ASCII)	M	32	2	

1. Compressor ==>>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

* Summary of SC2 messages

3	depends on use of the function at SC 2
2.x	depends on model type or options, see table 3
1	always

Sigma Control 2										SAM		Message			
JSON-RPC	PROFIBUS/ PROFINET		Modbus**				Modbus	DeviceNet EtherNet/IP		Send/Receive		Message		Availability	Description / possible cause
			Input		Holding										
BDSnd	S7 DBX	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)		
xAM1	1.0	0	21	30022	277	40278	0	0	0	32	65.0	0001	Direction of rotation	1	The compressor drive motor is turning in the wrong direction.
xAM2	1.1	0	21	30022	277	40278	1	0	1	32	65.1	0002	Motor T⇞	2.8	Compressor drive motor overheated. (PTC or PT100)
xAM3	1.2	0	21	30022	277	40278	2	0	2	32	65.2	0003	pRV ⇞	1	TÜV inspection: The activating pressure of the pressure relief valve on the oil separator tank has been exceeded.
xAM4	1.3	0	21	30022	277	40278	3	0	3	32	65.3	0004	EMERGENCY STOP	1	EMERGENCY STOP control device actuated.
xAM5	1.4	0	21	30022	277	40278	4	0	4	32	65.4	0005	Oil separator T⇞	2.4	Maximum air temperature at the oil separator tank outlet is exceeded.
xAM6	1.5	0	21	30022	277	40278	5	0	5	32	65.5	0006	Overvoltage suppressor	2.O	Overvoltage at power supply.
xAM7	1.6	0	21	30022	277	40278	6	0	6	32	65.6	0007	Mains monitor	2.O	Power supply fault (separate network monitoring module)
xAM8	1.7	0	21	30022	277	40278	7	0	7	32	65.7	0008	Diagnostics group alarm	1	A diagnostic alarm occurred.
xAM9	0.0	0	21	30022	277	40278	8	1	0	32	64.0	*	Sigma Control T⇞	1	Permissible internal temperature for SIGMA CONTROL 2 MCS or IOM exceeded.
xAM10	0.1	0	21	30022	277	40278	9	1	1	32	64.1	0010	Blow-off protection ⇞	2.1	The activating pressure of the pressure relief valve on the oil separator tank has been exceeded.
xAM11	0.2	0	21	30022	277	40278	10	1	2	32	64.2	*	Oil-/air cooler fan Group alarm	2.9	Group alarm oil-/air cooler fan (overload, FC alarm)
xAM12	0.3	0	21	30022	277	40278	11	1	3	32	64.3	0012	Access doors	2.21	Door open / interlocked panel removed while the machine is running.
xAM13	0.4	0	21	30022	277	40278	12	1	4	32	64.4	0013	Compressor motor overload	1	Overload shut-down of the compressor drive motor.
xAM14	0.5	0	21	30022	277	40278	13	1	5	32	64.5	0016	Air cooler - Fan Overcurrent	2.11	Overload shut-down of the air cooler fan motor.
xAM15	0.6	0	21	30022	277	40278	14	1	6	32	64.6	0015	ADT ⇞	1	Maximum permissible airend discharge temperature (ADT) exceeded.
xAM16	0.7	0	21	30022	277	40278	15	1	7	32	64.7	0014 *	Oil cooler fan Group alarm	2.10	Group alarm oil cooler fan (overload, FC alarm)

1. Compressor ==>>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

3	depends on use of the function at SC 2
2.x	depends on model type or options, see table 3
1	always

* Summary of SC2 messages

Sigma Control 2 JSON-RP BDSnd	PROFIBUS/ PROFINET S7 DBX	PROFIBUS/ PROFINET S5 DW	Modbus** Input Protocol addres	Modbus** Input Register	Modbus** Holding Protocol addres	Modbus** Holding Register	Modbus S5 Bit	DeviceNet EtherNet/IP Register	DeviceNet EtherNet/IP Bit	SAM Send/Receive S5 DW	SAM Send/Receive S7 DBX	Message ID	Message Text (SC2 Display, except *)	Avalability	Description / possible cause
xAM17	3.0	1	22	30023	278	40279	0	2	0	33	67.0	0017	Safety shutdown ADT	1	Maximum permissible airend discharge temperature (ADT) exceeded - safety shutdown.
xAM18	3.1	1	22	30023	278	40279	1	2	1	33	67.1	0018	Interior fan Overcurrent	2.12	Overload shut-down of the interior fan motor.
xAM19	3.2	1	22	30023	278	40279	2	2	2	33	67.2	0019	Internal pressure pi ⇟	2.1	Internal pressure too low in idle mode.
xAM20	3.3	1	22	30023	278	40279	3	2	3	33	67.3		free		
xAM21	3.4	1	22	30023	278	40279	4	2	4	33	67.4	0021	Refrigeration dryer T⇟	2.T	Refrigeration dryer: Compressed air temperature too low.
xAM22	3.5	1	22	30023	278	40279	5	2	5	33	67.5	0022	Oil separator dp ⇞	2.1	Oil separator cartridge clogged.
xAM23	3.6	1	22	30023	278	40279	6	2	6	33	67.6		free		
xAM24	3.7	1	22	30023	278	40279	7	2	7	33	67.7		free		
xAM25	2.0	1	22	30023	278	40279	8	3	0	33	66.0	*	First IOM - Group alarm	1	Fault in IOM itself
xAM26	2.1	1	22	30023	278	40279	9	3	1	33	66.1	*	Second IOM - Group alarm	2.6	Fault in IOM itself
xAM27	2.2	1	22	30023	278	40279	10	3	2	33	66.2		free		
xAM28	2.3	1	22	30023	278	40279	11	3	3	33	66.3		free		
xAM29	2.4	1	22	30023	278	40279	12	3	4	33	66.4	*	Third IOM - Group alarm	2.C	Fault in IOM itself
xAM30	2.5	1	22	30023	278	40279	13	3	5	33	66.5		free		
xAM31	2.6	1	22	30023	278	40279	14	3	6	33	66.6	0031	Air filter dp ⇞	2.19	The air filter or the air filter cartridge is clogged and needs to be changed.
xAM32	2.7	1	22	30023	278	40279	15	3	7	33	66.7	0032	Oil pressure ⇞	2.V1	Vaccum: Oil pressure is too high

1. Compressor ==>>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

3	depends on use of the function at SC 2
2.x	depends on model type or options, see table 3
1	always

* Summary of SC2 messages

JSON-RPC	Sigma Control 2									SAM		Message		Availability	Description / possible cause
	PROFIBUS/ PROFINET		Modbus**				Modbus	DeviceNet EtherNet/IP		Send/Receive					
			Input		Holding										
BDSnd	S7 DBX	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)		
xAM33	5.0	2	23	30024	279	40280	0	4	0	34	69.0	0033	Oil pressure ⇟	2.V	Vaccum: Oil pressure is too low
xAM34	5.1	2	23	30024	279	40280	1	4	1	34	69.1	*	Mains-/Delta-/Redundancy contactor on?	1	Mains contactor, delta contactor or redundancy contactor does not close.
xAM35	5.2	2	23	30024	279	40280	2	4	2	34	69.2	0035	Cabinet fan I ⇞	2.13	Overload shut-down of the control cabinet fan motor.
xAM36	5.3	2	23	30024	279	40280	3	4	3	34	69.3		free		
xAM37	5.4	2	23	30024	279	40280	4	4	4	34	69.4		free		
xAM38	5.5	2	23	30024	279	40280	5	4	5	34	69.5	0038	PD temperature ⇟	2.3	Pressure discharge temperature too low.
xAM39	5.6	2	23	30024	279	40280	6	4	6	34	69.6	0039	PD temperature ⇞	2.3	Pressure discharge temperature too high.
xAM40	5.7	2	23	30024	279	40280	7	4	7	34	69.7	*	Mains-/Delta-/Redundancy contactor off?	1	Mains contactor, delta contactor or redundancy contactor does not open.
xAM41	4.0	2	23	30024	279	40280	8	5	0	34	68.0	0041	Mains voltage ⇟	1	Second power failure.
xAM42	4.1	2	23	30024	279	40280	9	5	1	34	68.1	0042	Back pressure stop	1	Back pressure in the oil separator tank caused by defective venting. (stop = Motor off)
xAM43	4.2	2	23	30024	279	40280	10	5	2	34	68.2	0043	ADT rise dT/dt ⇞	1	The rate of rise of the airend discharge temperature (ADT) is too fast.
xAM44	4.3	2	23	30024	279	40280	11	5	3	34	68.3	0044	No pressure buildup	2.1	The machine does not produce compressed air. The working pressure does not rise above 3.5 bar / 51 psi within the preset
xAM45	4.4	2	23	30024	279	40280	12	5	4	34	68.4		free		
xAM46	4.5	2	23	30024	279	40280	13	5	5	34	68.5	*	Compressor motor FC Group alarm	2.F	Fault in frequency converter for compressor motor
xAM47	4.6	2	23	30024	279	40280	14	5	6	34	68.6	*	USS or modbus communication error	2.F	RS485-FC: USS or Modbus communication error
xAM48	4.7	2	23	30024	279	40280	15	5	7	34	68.7	*	High-voltage cell	2.C	Fault in the high voltage cell.

1. Compressor ==>>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

* Summary of SC2 messages

Availability	
3	depends on use of the function at SC 2
2.x	depends on model type or options, see table 3
1	always

Sigma Control 2										SAM		Message			
JSON-RPC	PROFIBUS/ PROFINET		Modbus**				Modbus	DeviceNet EtherNet/IP		Send/Receive				Availability	Description / possible cause
			Input		Holding										
BDSnd	S7 DBX	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)		
xAM49	7.0	3	24	30025	280	40281	0	6	0	35	71.0		free		
xAM50	7.1	3	24	30025	280	40281	1	6	1	35	71.1	0050	Customer-provided power element	2.C	Fault in customer-provided power element
xAM51	7.2	3	24	30025	280	40281	2	6	2	35	71.2	0051	Aggregat A	2.14	HSD: Aggregate A alarm, blocking aggregate B.
xAM52	7.3	3	24	30025	280	40281	3	6	3	35	71.3	0052	Aggregat B	2.14	HSD: Aggregate B alarm, blocking aggregate A.
xAM53	7.4	3	24	30025	280	40281	4	6	4	35	71.4	0053	Condensate drain 2	2.O	The condensate drain 2 is defective.
xAM54	7.5	3	24	30025	280	40281	5	6	5	35	71.5		free		
xAM55	7.6	3	24	30025	280	40281	6	6	6	35	71.6		free		
xAM56	7.7	3	24	30025	280	40281	7	6	7	35	71.7	0056	RD condensate drain	2.T	Refrigeration dryer: The condensate drain is defective.
xAM57	6.0	3	24	30025	280	40281	8	7	0	35	70.0		free		
xAM58	6.1	3	24	30025	280	40281	9	7	1	35	70.1	0058	Condensate drain 1	2.15	The condensate drain is defective.1
xAM59	6.2	3	24	30025	280	40281	10	7	2	35	70.2	0059	Back pressure run	1	Drive belt or coupling broken. (run = motor running)
xAM60	6.3	3	24	30025	280	40281	11	7	3	35	70.3	0060	Softstart	2.C	Fault in the softstarter
xAM61	6.4	3	24	30025	280	40281	12	7	4	35	70.4		free		
xAM62	6.5	3	24	30025	280	40281	13	7	5	35	70.5	0062	Refrigeration dryer p ⇞	2.T	Refrigeration dryer: Pressure too high in the refrigerant circuit. Safety pressure switch tripped.
xAM63	6.6	3	24	30025	280	40281	14	7	6	35	70.6	0063	Refrigeration dryer p ⇟	2.16	Refrigeration dryer: Refrigerant lost; pressure in the refrigerant circuit too low. Inlet pressure switched tripped.
xAM64	6.7	3	24	30025	280	40281	15	7	7	35	70.7	0064	Refrigeration dryer compressor motor temperature ⇞	2.17	Refrigeration dryer: Motor temperature too high. Temperature switch tripped.

1. Compressor ==>>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

Availability	
3	depends on use of the function at SC 2
2.x	depends on model type or options, see table 3
1	always

* Summary of SC2 messages

Sigma Control 2 JSON-RPC BDSnd	PROFIBUS/ PROFINET S7 DBX	PROFIBUS/ PROFINET S5 DW	Modbus** Input Protocol addres	Modbus** Input Register	Modbus** Holding Protocol addres	Modbus** Holding Register	Modbus S5 Bit	DeviceNet EtherNet/IP Register	DeviceNet EtherNet/IP Bit	SAM Send/Receive S5 DW	SAM Send/Receive S7 DBX	Message ID	Message Text (SC2 Display, except *)	Avalability	Description / possible cause
xAM65	9.0	4	25	30026	281	40282	0	8	0	36	73.0		free		
xAM66	9.1	4	25	30026	281	40282	1	8	1	36	73.1		free		
xAM67	9.2	4	25	30026	281	40282	2	8	2	36	73.2	0067	SC2 <=> SC2 communication error	2.14	HSD without SFC: Ethernet communication with corresponding aggregat is interrupted.
xAM68	9.3	4	25	30026	281	40282	3	8	3	36	73.3		free		
xAM69	9.4	4	25	30026	281	40282	4	8	4	36	73.4		free		
xAM70	9.5	4	25	30026	281	40282	5	8	5	36	73.5		free		
xAM71	9.6	4	25	30026	281	40282	6	8	6	36	73.6		free		
xAM72	9.7	4	25	30026	281	40282	7	8	7	36	73.7		free		
xAM73	8.0	4	25	30026	281	40282	8	9	0	36	72.0	0073	External message 1	3	External binary message 1
xAM74	8.1	4	25	30026	281	40282	9	9	1	36	72.1	0074	External message 2	3	External binary message 2
xAM75	8.2	4	25	30026	281	40282	10	9	2	36	72.2	0075	External message 3	3	External binary message 3
xAM76	8.3	4	25	30026	281	40282	11	9	3	36	72.3	0076	External message 4	3	External binary message 4
xAM77	8.4	4	25	30026	281	40282	12	9	4	36	72.4	0077	External message 5	3	External binary message 5
xAM78	8.5	4	25	30026	281	40282	13	9	5	36	72.5	0078	External message 6	3	External binary message 6
xAM79	8.6	4	25	30026	281	40282	14	9	6	36	72.6	0095	p-Switch pN	3	Adjustable pN-pressure switch
xAM80	8.7	4	25	30026	281	40282	15	9	7	36	72.7	0094	T-Switch ADT	3	Adjustable airend discharge temperature switch

1. Compressor ==>>>> Master Controller

Table 1.2 : Warning Messages and Maintenance Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

W	Warning	3	depends on use of the function at SC 2
M	Maintenance	2.x	depends on model type or options, see table 3
*	Summary of SC2 messages	1	always

Sigma Control 2										SAM		Message			
JSON-RPC	PROFIBUS/ PROFINET		Modbus**				Modbus	DeviceNet EtherNet/IP		Send/Receive		Message		Availability	Description / possible cause
			Input		Holding										
BDSnd	S7	S5	Protocol	Input	Protocol	Holding	S5			S5	S7				
	DBX	DW	addres	Register	addres	Register	Bit	Register	Bit	DW	DBX	ID	Text (SC2 Display, except *)		
xWM1	11.0	5	26	30027	282	40283	0	10	0	37	75.0	W 0001	Equipment number incomplete	1	The compressor equipment number consists of less than 7 characters.
xWM2	11.1	5	26	30027	282	40283	1	10	1	37	75.1	W 0002	Motor T↑	2.8	Drive motor overheating. (PTC oder PT100)
xWM3	11.2	5	26	30027	282	40283	2	10	2	37	75.2	M 0003	Air filter dp↑ (Prewarning)	2.26	The pre-filter of the air filter clogged or the cyclone and housing are dirty. Replace the pre-filter or clean the cyclon and housing.
xWM4	11.3	5	26	30027	282	40283	3	10	3	37	75.3	M 0004	Oil separator dp↑	2.1	The pressure drop across the oil separator cartridge has risen. Oil separator cartridge clogged.
xWM5	11.4	5	26	30027	282	40283	4	10	4	37	75.4	W 0005	Idle warming	3	The compressor is warming up in idle
xWM6	11.5	5	26	30027	282	40283	5	10	5	37	75.5	W 0006	Overvoltage suppressor	2.O	Overvoltage at power supply.
xWM7	11.6	5	26	30027	282	40283	6	10	6	37	75.6	W 0007	Mains monitor	2.O	Power supply fault (separate network monitoring module)
xWM8	11.7	5	26	30027	282	40283	7	10	7	37	75.7	W 0008	ADT ↑	1	Maximum airend discharge temperature will soon be reached.
xWM9	10.0	5	26	30027	282	40283	8	11	0	37	74.0	W	free		
xWM10	10.1	5	26	30027	282	40283	9	11	1	37	74.1	M 0010	Oilfilter 2 dp↑	2.25	The pressure differential of the 2nd oil filter has risen. Oil filter clogged.
xWM11	10.2	5	26	30027	282	40283	10	11	2	37	74.2	M 0011	Oilfilter 1 dp↑	2.18	The pressure differential of the 1st oil filter has risen. Oil filter clogged.
xWM12	10.3	5	26	30027	282	40283	11	11	3	37	74.3	W	free		
xWM13	10.4	5	26	30027	282	40283	12	11	4	37	74.4	M 0013	Air filter dp↑	2.19	The air filter or the air filter cartridge is clogged and needs to be changed.
xWM14	10.5	5	26	30027	282	40283	13	11	5	37	74.5	W	free		
xWM15	10.6	5	26	30027	282	40283	14	11	6	37	74.6	W 0015	Bus alarm	1	The bus link from the Com-Modul interface is interrupted.
xWM16	10.7	5	26	30027	282	40283	15	11	7	37	74.7	W	free		

1. Compressor ==>>>> Master Controller

Table 1.2 : Warning Messages and Maintenance Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

W	Warning
M	Maintenance
*	Summary of SC2 messages

3	depends on use of the function at SC 2
2.x	depends on model type or options, see table 3
1	always

Sigma Control 2										SAM		Message			
JSON-RPC	PROFIBUS/ PROFINET		Modbus**				Modbus	DeviceNet EtherNet/IP		Send/Receive				Availability	Description / possible cause
			Input		Holding										
BDSnd	S7 DBX	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)		
xWM17	13.0	6	27	30028	283	40284	0	12	0	38	77.0	W	free		
xWM18	13.1	6	27	30028	283	40284	1	12	1	38	77.1	W	free		
xWM19	13.2	6	27	30028	283	40284	2	12	2	38	77.2	W 0065	Refrigeration dryer external shut-off	2.T	Refrigeration dryer: Dryer shut-off via external signal.
xWM20	13.3	6	27	30028	283	40284	3	12	3	38	77.3	W 0064	Refrigeration dryer com-pressor motor temperature ‡	2.17	Refrigeration dryer: Motor temperature too high. Temperature switch tripped.
xWM21	13.4	6	27	30028	283	40284	4	12	4	38	77.4	W 0061	Refrigeration dryer T ‡	2.T	Refrigeration dryer: Compressed air temperature too low.
xWM22	13.5	6	27	30028	283	40284	5	12	5	38	77.5	W 0062	Refrigeration dryer p ‡	2.T	Refrigeration dryer: Pressure too high in the refrigerant circuit. Safety pressure switch tripped.
xWM23	13.6	6	27	30028	283	40284	6	12	6	38	77.6	W 0063	Refrigeration dryer p ‡	2.16	Refrigeration dryer: Refrigerant lost; pressure in the refrigerant circuit too low. Inlet pressure switched tripped.
xWM24	13.7	6	27	30028	283	40284	7	12	7	38	77.7	M 0024	Mains contactor operations ‡	1	Maximum switching operations of mains contactor exceeded.
xWM25	12.0	6	27	30028	283	40284	8	13	0	38	76.0	M 0025	Oil separator h ‡	1	Oil separator cartridge: Maintenance interval has elapsed.
xWM26	12.1	6	27	30028	283	40284	9	13	1	38	76.1	M 0026	Oil change h ‡	1	Cooling oil: Maintenance interval has elapsed.
xWM27	12.2	6	27	30028	283	40284	10	13	2	38	76.2	M 0027	Oil filter h ‡	1	Oil filter: Maintenance interval has elapsed.
xWM28	12.3	6	27	30028	283	40284	11	13	3	38	76.3	M 0028	Air filter h ‡	1	Air filter: Maintenance interval has elapsed.
xWM29	12.4	6	27	30028	283	40284	12	13	4	38	76.4	M 0029	Valve inspection h ‡	1	Valves: Maintenance interval has elapsed.
xWM30	12.5	6	27	30028	283	40284	13	13	5	38	76.5	M 0030	Belt/coupling inspection h ‡	1	Belt tension/coupling: Maintenance interval has elapsed.
xWM31	12.6	6	27	30028	283	40284	14	13	6	38	76.6	M 0031	Motor bearings h ‡	1	Motor bearing of compressor motor: Maintenance interval has elapsed.
xWM32	12.7	6	27	30028	283	40284	15	13	7	38	76.7	M 0032	Electrical equipment h ‡	1	Electric components and installation: Maintenance interval has elapsed.

1. Compressor ==>>>> Master Controller

Table 1.2 : Warning Messages and Maintenance Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

W	Warning	3	depends on use of the function at SC 2
M	Maintenance	2.x	depends on model type or options, see table 3
*	Summary of SC2 messages	1	always

Sigma Control 2											SAM		Message			
JSON-RPC	PROFIBUS/ PROFINET		Modbus**				Modbus	DeviceNet EtherNet/IP			Send/Receive				Availability	Description / possible cause
			Input		Holding											
BDSnd	S7	S5	Protocol	Input	Protocol	Holding	S5				S5	S7				
	DBX	DW	addres	Register	addres	Register	Bit	Register	Bit		DW	DBX	ID	Text (SC2 Display, except *)		
xWM33	15.0	7	28	30029	284	40285	0	14	0		39	79.0	M 0033	Fan bearings h ⇞	2.20	Motor bearing of fan motors: Maintenance interval has elapsed.
xWM34	15.1	7	28	30029	284	40285	1	14	1		39	79.1	W 0034	PD temperature ↓	2.3	Pressure discharge temperature low.
xWM35	15.2	7	28	30029	284	40285	2	14	2		39	79.2	W 0035	PD temperature ↑	2.3	Pressure discharge temperature high.
xWM36	15.3	7	28	30029	284	40285	3	14	3		39	79.3	W 0036	Motor starts /h ⇞	1	The permissible number of motor starts was exceeded in the last 60 minutes.
xWM37	15.4	7	28	30029	284	40285	4	14	4		39	79.4	W 0037	Motor starts /d ⇞	1	The permissible number of motor starts was exceeded in the last 24 hours.
xWM38	15.5	7	28	30029	284	40285	5	14	5		39	79.5	W 0038	Blow-off protection ↑	2.1	The pressure relief valve's activating pressure will soon be reached.
xWM39	15.6	7	28	30029	284	40285	6	14	6		39	79.6	W	free		
xWM40	15.7	7	28	30029	284	40285	7	14	7		39	79.7	M	free		
xWM41	14.0	7	28	30029	284	40285	8	15	0		39	78.0	W 0041	Mains voltage ↓	1	First power failure: The machine is automatically restarted.
xWM42	14.1	7	28	30029	284	40285	9	15	1		39	78.1	W	free		
xWM43	14.2	7	28	30029	284	40285	10	15	2		39	78.2	W 0043	External load signal?	3	Ambiguous external load signal: Increased cut-out pressure exceeded. The external load control has not switched to idle (off
xWM44	14.3	7	28	30029	284	40285	11	15	3		39	78.3	W 0044	Oil T↓	2.C	Cooling oil temperature too low.
xWM45	14.4	7	28	30029	284	40285	12	15	4		39	78.4	W 0045	DO test active	3	The DO test is active
xWM46	14.5	7	28	30029	284	40285	13	15	5		39	78.5	W 0046	System pressure ↓	3	Network pressure has fallen below the set 'low' value. Air consumption too high.
xWM47	14.6	7	28	30029	284	40285	14	15	6		39	78.6	W 0047	No pressure buildup	2.1	The compressor cannot build-up to working pressure.
xWM48	14.7	7	28	30029	284	40285	15	15	7		39	78.7	M 0048	Bearing lube h ⇞	1	Re-grease the motor bearings. Maintenance interval has elapsed.

1. Compressor ==>>>> Master Controller

Table 1.2 : Warning Messages and Maintenance Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

W	Warning
M	Maintenance
*	Summary of SC2 messages

3	depends on use of the function at SC 2
2.x	depends on model type or options, see table 3
1	always

Sigma Control 2										SAM		Message		Avalability	Description / possible cause
JSON-RPC	PROFIBUS/ PROFINET		Modbus**				Modbus	DeviceNet EtherNet/IP		Send/Receive					
			Input		Holding										
BDSnd	S7 DBX	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)		
xWM49	17.0	8	29	30030	285	40286	0	16	0	40	81.0	M 0049	Annual maintenance	1	Appears one year after last maintenance.
xWM50	17.1	8	29	30030	285	40286	1	16	1	40	81.1	M 0050	Fan bearing lube h ⇞	2.22	Fan bearing lube: Maintenance interval has elapsed.
xWM51	17.2	8	29	30030	285	40286	2	16	2	40	81.2	W 0051	Double aggregate emergency operation!	2.14	HSD: Aggregate is in single run mode.
xWM52	17.3	8	29	30030	285	40286	3	16	3	40	81.3	W	free		
xWM53	17.4	8	29	30030	285	40286	4	16	4	40	81.4	W 0053	Condensate drain 2	2.O	The condensate drain 2 is defective.
xWM54	17.5	8	29	30030	285	40286	5	16	5	40	81.5	W	free		
xWM55	17.6	8	29	30030	285	40286	6	16	6	40	81.6	M 0055	External counter 1	3	Freely definable maintenance hours counter 1: Maintenance interval has elapsed
xWM56	17.7	8	29	30030	285	40286	7	16	7	40	81.7	M 0056	External counter 2	3	Freely definable maintenance hours counter 2: Maintenance interval has elapsed
xWM57	16.0	8	29	30030	285	40286	8	17	0	40	80.0	M 0057	External counter 3	3	Freely definable maintenance hours counter 3: Maintenance interval has elapsed
xWM58	16.1	8	29	30030	285	40286	9	17	1	40	80.1	W 0058	SC2 <=> SC2 Communication error	2.14	SC2 to SC2 communication is interrupted.
xWM59	16.2	8	29	30030	285	40286	10	17	2	40	80.2	W 0059	Start T↓↓	1	The airend temperature is too low (<−10°C / 14°F) for the machine to be operated.
xWM60	16.3	8	29	30030	285	40286	11	17	3	40	80.3	W 0060	Start T↓	1	The airend temperature is too low (<+2°C / 36°F).
xWM61	16.4	8	29	30030	285	40286	12	17	4	40	80.4	W 0061	Compressor T ↓	1	Compressor temperature (ADT or OST) did not reach the minimum required value during operation.
xWM62	16.5	8	29	30030	285	40286	13	17	5	40	80.5	W *	Compressor motor FC Service mode active	2.F	Service mode of frequency converter is active. FC can be operated for test purposes.
xWM63	16.6	8	29	30030	285	40286	14	17	6	40	80.6	W 0063	Compressor motor FC test shut-off required	2.F	Test of "Safe Torque off" shut down at frequency converter.
xWM64	16.7	8	29	30030	285	40286	15	17	7	40	80.7	W *	Compressor motor FC error AI	2.F	Fault at AI0 (Sinamics) or AI1 (Masterdrive) of the frequency converter.

1. Compressor ==>>>> Master Controller

Table 1.2 : Warning Messages and Maintenance Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

W	Warning
M	Maintenance
*	Summary of SC2 messages

3	depends on use of the function at SC 2
2.x	depends on model type or options, see table 3
1	always

Sigma Control 2											SAM		Message			
JSON-RPC	PROFIBUS/ PROFINET		Modbus**				Modbus	DeviceNet EtherNet/IP		Send/Receive					Avalability	Description / possible cause
			Input		Holding											
BDSnd	S7	S5	Protocol addres	Input Register	Protocol addres	Holding Register	S5			S5	S7					
	DBX	DW					Bit	Register	Bit	DW	DBX	ID	Text (SC2 Display, except *)			
xWM65	19.0	9	30	30031	286	40287	0	18	0	41	83.0	W *	Compressor motor FC Group aarm		2.F	The compressor motor was restarted automatically after a fault occurred twice on the frequency converter.
xWM66	19.1	9	30	30031	286	40287	1	18	1	41	83.1	W				
xWM67	19.2	9	30	30031	286	40287	2	18	2	41	83.2	W 0067	System pressure ↑		2.V	Vacuum: Network pressure has risen above the set 'high' value. Air suction too low.
xWM68	19.3	9	30	30031	286	40287	3	18	3	41	83.3	W 0068	Condensate drain 1		2.1 5	The condensate drain 1 is defective.
xWM69	19.4	9	30	30031	286	40287	4	18	4	41	83.4	W 0069	Error operation without RD → Call service!		2.T	Error operation without refrigeration dryer is activ. The compressor is running without dryer for a certain time.
xWM70	19.5	9	30	30031	286	40287	5	18	5	41	83.5	W 0070	Refrigeration dryer T↑		2.T	Refrigeration dryer: Compressed air temperature too high.
xWM71	19.6	9	30	30031	286	40287	6	18	6	41	83.6	M 0071	Oil level ↓		2.C	Cooling oil level too low.
xWM72	19.7	9	30	30031	286	40287	7	18	7	41	83.7	W 0072	RD condensate drain		2.T	Refrigeration dryer: The condensate drain is defective.
xWM73	18.0	9	30	30031	286	40287	8	19	0	41	82.0	W 0073	External message 1		3	External binary message 1
xWM74	18.1	9	30	30031	286	40287	9	19	1	41	82.1	W 0074	External message 2		3	External binary message 2
xWM75	18.2	9	30	30031	286	40287	10	19	2	41	82.2	W 0075	External message 3		3	External binary message 3
xWM76	18.3	9	30	30031	286	40287	11	19	3	41	82.3	W 0076	External message 4		3	External binary message 4
xWM77	18.4	9	30	30031	286	40287	12	19	4	41	82.4	W 0077	External message 5		3	External binary message 5
xWM78	18.5	9	30	30031	286	40287	13	19	5	41	82.5	W 0078	External message 6		3	External binary message 6
xWM79	18.6	9	30	30031	286	40287	14	19	6	41	82.6	W 0095	p-Switch pN		3	Adjustable pN-pressure switch
xWM80	18.7	9	30	30031	286	40287	15	19	7	41	82.7	W 0094	T-Switch ADT		3	Adjustable airend discharge temperature switch

1. Compressor ==>>>> Master Controller

Table 1.3 : Operational messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

** SXC

* Summary of SC2 messages

3	depends on use of the function at SC 2
2.x	depends on model type or options, see table 3
1	always

Sigma Control 2										SAM		Message		Availability	Description / possible cause
JSON-RPC	PROFIBUS/ PROFINET		Modbus**				Modbus	DeviceNet EtherNet/IP		Send/Receive					
			Input		Holding										
BDSnd	S7 DBX	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)		
xOM1	21.0	10	31	30032	287	40288	0	20	0	42	85.0	0001	Delayed start (…s) activ	3	The compressor starts …s delayed after power on.
xOM2	21.1	10	31	30032	287	40288	1	20	1	42	85.1	0002	Technican on site	1	A technichan works at the machine
xOM3	21.2	10	31	30032	287	40288	2	20	2	42	85.2		free		
xOM4	21.3	10	31	30032	287	40288	3	20	3	42	85.3		free		
xOM5	21.4	10	31	30032	287	40288	4	20	4	42	85.4		free		
xOM6	21.5	10	31	30032	287	40288	5	20	5	42	85.5		free		
xOM7	21.6	10	31	30032	287	40288	6	20	6	42	85.6		free		
xOM8	21.7	10	31	30032	287	40288	7	20	7	42	85.7		free		
xOM9	20.0	10	31	30032	287	40288	8	21	0	42	84.0		free		
xOM10	20.1	10	31	30032	287	40288	9	21	1	42	84.1		free		
xOM11	20.2	10	31	30032	287	40288	10	21	2	42	84.2	0011	Cold start release	3	Cold start is enabled
xOM12	20.3	10	31	30032	287	40288	11	21	3	42	84.3		free		
xOM13	20.4	10	31	30032	287	40288	12	21	4	42	84.4		free		
xOM14	20.5	10	31	30032	287	40288	13	21	5	42	84.5		free		
xOM15	20.6	10	31	30032	287	40288	14	21	6	42	84.6	0020/ 0021**	Refrigeration dryer out of action!	2.T	Machine running in "Error operation without Refrigeration dryer" operating mode
xOM16	20.7	10	31	30032	287	40288	15	21	7	42	84.7		free		

1. Compressor ==>>>> Master Controller

Table 1.3 : Operational messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

** SXC * Summary of SC2 messages

Availability	
3	depends on use of the function at SC 2
2.x	depends on model type or options, see table 3
1	always

Sigma Control 2 JSON-RPC BDSnd	PROFIBUS/ PROFINET S7 DBX	PROFIBUS/ PROFINET S5 DW	Modbus** Input Protocol addres	Modbus** Input Register	Modbus** Holding Protocol addres	Modbus** Holding Register	Modbus S5 Bit	DeviceNet EtherNet/IP Register	DeviceNet EtherNet/IP Bit	SAM Send/Receive S5 DW	SAM Send/Receive S7 DBX	Message ID	Message Text (SC2 Display, except *)	Avalability	Description / possible cause
xOM17	23.0	11	32	30033	288	40289	0	22	0	43	87.0	0073	External message 1	3	External binary message 1
xOM18	23.1	11	32	30033	288	40289	1	22	1	43	87.1	0074	External message 2	3	External binary message 2
xOM19	23.2	11	32	30033	288	40289	2	22	2	43	87.2	0075	External message 3	3	External binary message 3
xOM20	23.3	11	32	30033	288	40289	3	22	3	43	87.3	0076	External message 4	3	External binary message 4
xOM21	23.4	11	32	30033	288	40289	4	22	4	43	87.4	0077	External message 5	3	External binary message 5
xOM22	23.5	11	32	30033	288	40289	5	22	5	43	87.5	0078	External message 6	3	External binary message 6
xOM23	23.6	11	32	30033	288	40289	6	22	6	43	87.6	0095	p-Switch pN	3	Adjustable pN-pressure switch
xOM24	23.7	11	32	30033	288	40289	7	22	7	43	87.7	0094	T-Switch ADT	3	Adjustable airend discharge temperature switch
xOM25	22.0	11	32	30033	288	40289	8	23	0	43	86.0		free		
xOM26	22.1	11	32	30033	288	40289	9	23	1	43	86.1		free		
xOM27	22.2	11	32	30033	288	40289	10	23	2	43	86.2	0027	Power OFF → ON	1	It is necessary to switch power supply of SC2 off and on.
xOM28	22.3	11	32	30033	288	40289	11	23	3	43	86.3	0028	Dynamic motor T↑	2.5	Control mode DYNAMIC: The temperature of the compressor motor is too high.
xOM29	22.4	11	32	30033	288	40289	12	23	4	43	86.4		free		
xOM30	22.5	11	32	30033	288	40289	13	23	5	43	86.5	0030	Voltage restored	1	SC2 has been feed with supply voltage.
xOM31	22.6	11	32	30033	288	40289	14	23	6	43	86.6		free		
xOM32	22.7	11	32	30033	288	40289	15	23	7	43	86.7		free		

1. Compressor ==>>>> Master Controller

Table 1.3 : Operational messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

** SXC

3	depends on use of the function at SC 2
2.x	depends on model type or options, see table 3
1	always

* Summary of SC2 messages

JSON-RPC	Sigma Control 2 PROFIBUS/ PROFINET		Modbus** Input		Holding		Modbus	DeviceNet EtherNet/IP		SAM Send/Receive		Message		Availability	Description / possible cause
BDSnd	S7 DBX	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)		
xOM33	25.0	12	33	30034	289	40290	0	24	0	44	89.0	0033	Machine report	3	Life sign has been sent.
xOM34	25.1	12	33	30034	289	40290	1	24	1	44	89.1		free		
xOM35	25.2	12	33	30034	289	40290	2	24	2	44	89.2		free		
xOM36	25.3	12	33	30034	289	40290	3	24	3	44	89.3		free		
xOM37	25.4	12	33	30034	289	40290	4	24	4	44	89.4		free		
xOM38	25.5	12	33	30034	289	40290	5	24	5	44	89.5	*	Safety shutdown …	1	Safety shutdown by safety part of SC2 software occured.
xOM39	25.6	12	33	30034	289	40290	6	24	6	44	89.6	*	ADT channel error … detected	1	Temporary accidental ground or open circuit at AIR1.00 (without safety shutdown)
xOM40	25.7	12	33	30034	289	40290	7	24	7	44	89.7	*	IOSlot Undervoltage error	1	Undervoltage error at at least one IOM.

1. Compressor ==>>>> Master Controller

Table 1.4 : Binary Signals

Availability	
3	depends on use of the function
2.x	see table 3
1	always

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

Sigma Control 2										SAM					
JSON-RP	PROFIBUS/ PROFINET		Modbus** Input		Modbus** Holding		Modbus	DeviceNet EtherNet/IP		Send/Receive		Description	Availability	Signal	
BDSnd	S7 DBX	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			"1"	"0"
xBS0	27.0	13	34	30035	290	40291	0	26	0	45	91.0	Controller on (LED at key "I" on) + Remote on + Bus (Receive)	1	yes	no
xBS1	27.1	13	34	30035	290	40291	1	26	1	45	91.1	Motor running	1	yes	no
xBS2	27.2	13	34	30035	290	40291	2	26	2	45	91.2	Load run	1	yes	no
xBS3	27.3	13	34	30035	290	40291	3	26	3	45	91.3	Idle	1	yes	no
xBS4	27.4	13	34	30035	290	40291	4	26	4	45	91.4	Local mode (LED at key "Remote" off)	1	yes	no
xBS5	27.5	13	34	30035	290	40291	5	26	5	45	91.5	Controller on (LED at key "I" on)	1	yes	no
xBS6	27.6	13	34	30035	290	40291	6	26	6	45	91.6	Group alarm	1	yes	no
xBS7	27.7	13	34	30035	290	40291	7	26	7	45	91.7	Group warning or group maintenance	1	yes	no
xBS8	26.0	13	34	30035	290	40291	8	27	0	45	90.0	Compressor ready for load	1	yes	no
xBS9	26.1	13	34	30035	290	40291	9	27	1	45	90.1	Compressor ON (Motor is running or ready for operation)	1	yes	no
xBS10	26.2	13	34	30035	290	40291	10	27	2	45	90.2	Oil-/air cooler fan is running	2.9	yes	no
xBS11	26.3	13	34	30035	290	40291	11	27	3	45	90.3	Oilcooler fan is running	2.10	yes	no
xBS12	26.4	13	34	30035	290	40291	12	27	4	45	90.4	Air cooler is fan running	2.11	yes	no
xBS13	26.5	13	34	30035	290	40291	13	27	5	45	90.5	Interior fan is running	2.12	yes	no
xBS14	26.6	13	34	30035	290	40291	14	27	6	45	90.6	Group maintenance (maintenance messages only)	1	yes	no
xBS15	26.7	13	34	30035	290	40291	15	27	7	45	90.7	Group warning (warning messages only)	1	yes	no

1. Compressor ==>>>> Master Controller

Table 1.4 : Binary Signals

3	depends on use of the function
2.x	see table 3
1	always

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

Sigma Control 2											SAM		Description	Availability	Signal	
JSON-RPC	PROFIBUS/ PROFINET		Modbus** Input		Modbus** Holding		Modbus	DeviceNet EtherNet/IP			Send/Receive					
BDSnd	S7 DBX	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	S5 Bit	Register	Bit		S5 DW	S7 DBX			"1"	"0"
xBS16	29.0	14	35	30036	291	40292	0	28	0		46	93.0	Mode *compressor ON from "I" key only* is active	3	yes	no
xBS17	29.1	14	35	30036	291	40292	1	28	1		46	93.1	Mode *compressor ON from "I" key and clock* is active	3	yes	no
xBS18	29.2	14	35	30036	291	40292	2	28	2		46	93.2				
xBS19	29.3	14	35	30036	291	40292	3	28	3		46	93.3				
xBS20	29.4	14	35	30036	291	40292	4	28	4		46	93.4	Mode *compressor ON from "I" key and external contact* is active	3	yes	no
xBS21	29.5	14	35	30036	291	40292	5	28	5		46	93.5	Mode *compressor ON from "I" key and clock or external contact* is active	3	yes	no
xBS22	29.6	14	35	30036	291	40292	6	28	6		46	93.6	Mode *compressor ON from "I" key and bus master* is active	3	yes	no
xBS23	29.7	14	35	30036	291	40292	7	28	7		46	93.7	Mode compressor ON (xBS16..xBS22) set from	3	bus master	local
xBS24	28.0	14	35	30036	291	40292	8	29	0		46	92.0	DUAL control mode is active	3	yes	no
xBS25	28.1	14	35	30036	291	40292	9	29	1		46	92.1	QUADRO control mode is active	3	yes	no
xBS26	28.2	14	35	30036	291	40292	10	29	2		46	92.2	VARIO control mode is active	3	yes	no
xBS27	28.3	14	35	30036	291	40292	11	29	3		46	92.3	DYNAMIC control mode is active	2.5	yes	no
xBS28	28.4	14	35	30036	291	40292	12	29	4		46	92.4	Continuous control mode is active	3	yes	no
xBS29	28.5	14	35	30036	291	40292	13	29	5		46	92.5				
xBS30	28.6	14	35	30036	291	40292	14	29	6		46	92.6				
xBS31	28.7	14	35	30036	291	40292	15	29	7		46	92.7	Control mode (xBS24 - xBS28) set from	3	bus master	local

1. Compressor ==>>>> Master Controller

Table 1.4 : Binary Signals

3	depends on use of the function
2.x	see table 3
1	always

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

Sigma Control 2											SAM		Description	Avalability	Signal	
JSON-RPC	PROFIBUS/ PROFINET		Modbus**				Modbus	DeviceNet EtherNet/IP			Send/Receive					
			Input		Holding											
BDSnd	S7 DBX	S5 DW	Protocol addres	Input Register	Protocol addres	Holding Register	S5 Bit	Register	Bit		S5 DW	S7 DBX			"1"	"0"
xBS32	31.0	15	36	30037	292	40293	0	30	0		47	95.0				
xBS33	31.1	15	36	30037	292	40293	1	30	1		47	95.1				
xBS34	31.2	15	36	30037	292	40293	2	30	2		47	95.2				
xBS35	31.3	15	36	30037	292	40293	3	30	3		47	95.3				
xBS36	31.4	15	36	30037	292	40293	4	30	4		47	95.4	Mode *load-idle signal from external contact* is active	3	yes	no
xBS37	31.5	15	36	30037	292	40293	5	30	5		47	95.5				
xBS38	31.6	15	36	30037	292	40293	6	30	6		47	95.6	Mode *load-idle signal from bus master* is active	3	yes	no
xBS39	31.7	15	36	30037	292	40293	7	30	7		47	95.7	Mode load-idle signal (xBS32..xBS47) set from	3	bus master	local
xBS40	30.0	15	36	30037	292	40293	8	31	0		47	94.0	Mode *load-idle signal from pressure regualtor, pA permanent* is active	3	yes	no
xBS41	30.1	15	36	30037	292	40293	9	31	1		47	94.1	Mode *load-idle signal from pressure regualtor, pB permanent* is active	3	yes	no
xBS42	30.2	15	36	30037	292	40293	10	31	2		47	94.2	Mode *load-idle signal from pressure regualtor, pA/pB change via clock* is active	3	yes	no
xBS43	30.3	15	36	30037	292	40293	11	31	3		47	94.3	Mode *load-idle signal from pressure regualtor, pA/pB change via timer* is active	3	yes	no
xBS44	30.4	15	36	30037	292	40293	12	31	4		47	94.4				
xBS45	30.5	15	36	30037	292	40293	13	31	5		47	94.5	Mode *load-idle signal from pressure regualtor, pA/pB change via external contact* is active	3	yes	no
xBS46	30.6	15	36	30037	292	40293	14	31	6		47	94.6	Mode *load-idle signal from pressure regualtor, pA/pB change via SC2 master (Ethernet)* is active	3	yes	no
xBS47	30.7	15	36	30037	292	40293	15	31	7		47	94.7				

1. Compressor ==>>>> Master Controller

Table 1.4 : Binary Signals

3	depends on use of the function
2.x	see table 3
1	always

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

JSON-RPC BDSnd	Sigma Control 2 PROFIBUS/PROFINET S7 DBX	PROFIBUS/PROFINET S5 DW	Modbus** Input Protocol addres	Modbus** Input Register	Modbus** Holding Protocol addres	Modbus** Holding Register	Modbus S5 Bit	DeviceNet EtherNet/IP Register	DeviceNet EtherNet/IP Bit	SAM Send/Receive S5 DW	SAM Send/Receive S7 DBX	Description	Avalability	Signal "1"	Signal "0"
xBS48	33.0	16	37	30038	293	40294	0	32	0	48	97.0	Idle warming is currently running or will start after next motor start	3	yes	no
xBS49	33.1	16	37	30038	293	40294	1	32	1	48	97.1				
xBS50	33.2	16	37	30038	293	40294	2	32	2	48	97.2				
xBS51	33.3	16	37	30038	293	40294	3	32	3	48	97.3	Automatic restart after power on	1	activated	deactivated
xBS52	33.4	16	37	30038	293	40294	4	32	4	48	97.4				
xBS53	33.5	16	37	30038	293	40294	5	32	5	48	97.5				
xBS54	33.6	16	37	30038	293	40294	6	32	6	48	97.6				
xBS55	33.7	16	37	30038	293	40294	7	32	7	48	97.7	Message acknowledgement by bus master	1	activated	deactivated
xBS56	32.0	16	37	30038	293	40294	8	33	0	48	96.0				
xBS57	32.1	16	37	30038	293	40294	9	33	1	48	96.1				
xBS58	32.2	16	37	30038	293	40294	10	33	2	48	96.2				
xBS59	32.3	16	37	30038	293	40294	11	33	3	48	96.3				
xBS60	32.4	16	37	30038	293	40294	12	33	4	48	96.4				
xBS61	32.5	16	37	30038	293	40294	13	33	5	48	96.5				
xBS62	32.6	16	37	30038	293	40294	14	33	6	48	96.6				
xBS63	32.7	16	37	30038	293	40294	15	33	7	48	96.7				

2. Master Controller ==>>>> Compressor

* negative value
1) write both values at a time
2) write both values at a time

Usability	
3	see associated table (2.x
2.x	see table 3
1	always

Table 2: Process data

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

Sigma Control 2						SAM		Description		Usability	Unit	Format
JSON-RPC	PROFIBUS/ PROFINET		Modbus**		DeviceNet EtherNet/IP	Send/Receive						
BDRcv	S7 DBW	S5 DW	Protoco. address	Holding Register	Register MSB/LSB	S5 DW	S7 DBW					
	0	0	0	40001	1/0	0	0	Receive bit xBS15 - xBS0	Table 2.1	3	binary	BOOL
	2	1	1	40002	3/2	1	2	Receive bit xBS31 - xBS16	Table 2.1	3	binary	BOOL
	4	2	2	40003	5/4	2	4	Receive bit xBS47 - xBS32	Table 2.1	3	binary	BOOL
	6	3	3	40004	7/6	3	6	Receive bit xBS63 - xBS48	Table 2.1	3	binary	BOOL
iDV0	8	4	4	40005	9/8	4	8	Setpoint pressure pA switching point 1)	*if xBS68=1*	1	mbar	INT
iDV1	10	5	5	40006	11/10	5	10	Setpoint pressure pA switching difference * 1)	*if xBS68=1*	1	mbar	INT
iDV2	12	6	6	40007	13/12	6	12	Setpoint pressure pB switching point 2)	*if xBS70=1*	1	mbar	INT
iDV3	14	7	7	40008	15/14	7	14	Setpoint pressure pB switching difference * 2)	*if xBS70=1*	1	mbar	INT
iDV4	16	8	8	40009	17/16	8	16	Speed	*if xBS48=1*	2.23	n-1	INT
iDV5	18	9	9	40010	19/18	9	18					
iDV6	20	10	10	40011	21/20	10	20					
iDV7	22	11	11	40012	23/22	11	22					
iDV8	24	12	12	40013	25/24	12	24					
iDV9	26	13	13	40014	27/26	13	26	Actual air receiver system pressure	*if xBS77=1*	1	mbar	INT
iDV10	28	14	14	40015	29/28	14	28					
iDV11	30	15	15	40016	31/30	15	30					
iDV12	32	16	16	40017	33/32	16	32					
	34	17	17	40018	35/34	17	34	Receive bit xBS79 - xBS64	Table 2.1	3	binary	BOOL
	36	18	18	40019	37/36	18	36	Receive bit xBS95 - xBS80	Table 2.1	3	binary	BOOL
iMasterMsgHeader	38	19	19	40020	39/38	19	38	Message header master		1		WORD
iDV13	40	20	20	40021	41/40	20	40					
iDV14	42	21	21	40022	43/42	21	42					
iDV15	44	22	22	40023	45/44	22	44					

2. Master Controller ==>>>> Compressor

Table 2: Process data

* negative value
1) write both values at a time
2) write both values at a time
** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

3	see associated table (2.x)
2.x	see table 3
1	always

Sigma Control 2						SAM		Description	Usability	Unit	Format
JSON-RPC	PROFIBUS/ PROFINET		Modbus**		DeviceNet EtherNet/IP	Send/Receive					
BDRcv	S7 DBW	S5 DW	Protocol address	Holding Register	Register MSB/LSB	S5 DW	S7 DBW				
iDV16	46	23	23	40024	47/46	23	46				
iDV17	48	24	24	40025	49/48	24	48				
iDV18	50	25	25	40026	51/50	25	50				
iDV19	52	26	26	40027	53/52	26	52				
iDV20	54	27	27	40028	55/54	27	54				
iDV21	56	28	28	40029	57/56	28	56				
iDV22	58	29	29	40030	59/58	29	58				
iDV23	60	30	30	40031	61/60	30	60				
iMasterMsgFooter	62	31	31	40032	63/62	31	62	Message footer master	1		WORD

2. Master Controller ==>>>> Compressor

Table 2: Busparameter, Modbus only

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2 * not for Modbus TCP

Sigma Control 2							SAM		Description	Example		
JSON-RPC	PROFIBUS/ PROFINET		Modbus **			DeviceNe EtherNet/IP	Send/Receive					
BDSnd	S7 DBB	S5 DW	Protocol addres	Holding Register	Bit	Register MSB/LSB	S5 DW	S7 DBB		HE	Valu	Result
%	%	%	512	40513		%	%	%	Modbus: Slaveaddress	64	100	Slave: 100
%	%	%	513	40514		%	%	%	Modbus: Priority, stop bits, baudrate *	16		19200/even/1stop
									Even parity, 1 stop bit (default)			
									Odd parity, 1 stop bit			
									No parity, 2 stop bits			
									No parity, 1 stop bit			
			513	40514	0				0 1 0 1			
			513	40514	1				0 0 1 1			
									1200bit/s			
									2400bit/s			
									4800bit/s			
									9600bit/s			
									19200bit/s (default)			
									38400bit/s			
									57600bit/s			
									76800bit/s			
									115200bit/s			
			513	40514	2				0 1 0 1 0 1 0 1 0			
			513	40514	3				0 0 1 1 0 0 1 1 0			
			513	40514	4				0 0 0 0 1 1 1 1 0			
			513	40514	5				0 0 0 0 0 0 0 0 1			
			513	40514	6				reserved, set to 0			
			513	40514	7				reserved, set to 0			
%	%	%	514	40515		%	%	%	Modbus: Modus: 0=RTU, 1=ASCII *	0	RTU	Modus: RTU
%	%	%	515	40516		%	%	%	Modbus: Timeout in ms	7D0	2000	Timeout: 2000ms

2. Master Controller ==>>>> Compressor

Table 2.1 : Receive bit assignment

Usability	
2.x	depends on model type or options, see table 3
1	always

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

Sigma Control 2								SAM		Description	Usability	Signal		
JSON-RPC	PROFIBUS/ PROFINET		Modbus**		Modbus	DeviceNet EtherNet/IP		Send/Receive						
BDSnd	S7 DBX	S5 DW	Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			"1"	"0"	Kind
xBS0	1.0	0	0	40001	0	0	0	0	1.0	SAM connection	1	yes	no	duration
xBS1	1.1	0	0	40001	1	0	1	0	1.1	Fluid: not used Vacuum: *Load run* by bus master *(if xBS2 = 1)*	2.V	--- load run	--- idle	--- duration
xBS2	1.2	0	0	40001	2	0	2	0	1.2	Fluid: not used Vacuum: Mode *load-idle signal from bus master*	2.V	--- yes (load/idle	--- local	--- duration
xBS3	1.3	0	0	40001	3	0	3	0	1.3	Continue drive motor running *(if xBS0=1)*	1	continue running	no bearing	duration
xBS4	1.4	0	0	40001	4	0	4	0	1.4					
xBS5	1.5	0	0	40001	5	0	5	0	1.5					
xBS6	1.6	0	0	40001	6	0	6	0	1.6					
xBS7	1.7	0	0	40001	7	0	7	0	1.7					
xBS8	0.0	0	0	40001	8	1	0	0	0.0					
xBS9	0.1	0	0	40001	9	1	1	0	0.1					
xBS10	0.2	0	0	40001	10	1	2	0	0.2					
xBS11	0.3	0	0	40001	11	1	3	0	0.3					
xBS12	0.4	0	0	40001	12	1	4	0	0.4					
xBS13	0.5	0	0	40001	13	1	5	0	0.5					
xBS14	0.6	0	0	40001	14	1	6	0	0.6					
xBS15	0.7	0	0	40001	15	1	7	0	0.7					

2. Master Controller ==>>>> Compressor

Table 2.1 : Receive bit assignment

2.x	depends on model type or options, see table 3
1	always

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

Sigma Control 2								SAM		Description	Usability	Signal		
JSON-RPC	PROFIBUS/ PROFINET		Modbus**		Modbus	DeviceNet EtherNet/IP		Send/Receive						
BDSnd	S7 DBX	S5 DW	Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			"1"	"0"	Kind
xBS16	3.0	1	1	40002	0	2	0	1	3.0	Mode *compressor ON from "I" key only*	1	yes	local	duration
xBS17	3.1	1	1	40002	1	2	1	1	3.1	Mode *compressor ON from "I" key and clock*	1	yes	local	duration
xBS18	3.2	1	1	40002	2	2	2	1	3.2					
xBS19	3.3	1	1	40002	3	2	3	1	3.3					
xBS20	3.4	1	1	40002	4	2	4	1	3.4	Mode *compressor ON from "I" key and external contact*	1	yes	local	duration
xBS21	3.5	1	1	40002	5	2	5	1	3.5	Mode *compressor ON from "I" key and clock or external contact*	1	yes	local	duration
xBS22	3.6	1	1	40002	6	2	6	1	3.6	Mode *compressor ON from "I" key and bus master*	1	yes (ON/OFF via xBS23)	local	duration
xBS23	3.7	1	1	40002	7	2	7	1	3.7	*Compressor ON / OFF* by bus master *(if xBS22 = 1)*	1	ON	OFF	duration
xBS24	2.0	1	1	40002	8	3	0	1	2.0	DUAL control mode	1	DUAL	local if xBS24 and xBS25 and xBS26 and xBS27 and xBS28 is "0"	duration
xBS25	2.1	1	1	40002	9	3	1	1	2.1	QUADRO control mode	1	QUADRO		duration
xBS26	2.2	1	1	40002	10	3	2	1	2.2	VARIO control mode	1	VARIO		duration
xBS27	2.3	1	1	40002	11	3	3	1	2.3	DYNAMIC control mode	2.5	DYNAMIC		duration
xBS28	2.4	1	1	40002	12	3	4	1	2.4	Continuous control mode	1	Continuous		duration
xBS29	2.5	1	1	40002	13	3	5	1	2.5					
xBS30	2.6	1	1	40002	14	3	6	1	2.6					
xBS31	2.7	1	1	40002	15	3	7	1	2.7					

2. Master Controller ==>>>> Compressor

Table 2.1 : Receive bit assignment

Usability	
2.x	depends on model type or options, see table 3
1	always

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

Sigma Control 2								SAM						
JSON-RPC	PROFIBUS/ PROFINET		Modbus**		Modbus	DeviceNet EtherNet/IP		Send/Receive		Description	Usability	Signal		
BDSnd	S7 DBX	S5 DW	Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			"1"	"0"	Kind
xBS32	5.0	2	2	40003	0	4	0	2	5.0					
xBS33	5.1	2	2	40003	1	4	1	2	5.1					
xBS34	5.2	2	2	40003	2	4	2	2	5.2					
xBS35	5.3	2	2	40003	3	4	3	2	5.3					
xBS36	5.4	2	2	40003	4	4	4	2	5.4	Mode *load-idle signal from external contact*	1	yes	local	duration
xBS37	5.5	2	2	40003	5	4	5	2	5.5					
xBS38	5.6	2	2	40003	6	4	6	2	5.6	Mode *load-idle signal from bus master*	1	yes (load/idle via xBS39)	local	duration
xBS39	5.7	2	2	40003	7	4	7	2	5.7	*Load run* by bus master *(if xBS38 = 1)*	1	load run	idle	duration
xBS40	4.0	2	2	40003	8	5	0	2	4.0	Mode *load-idle signal from pressure regualtor, pA permanent*	1	yes	local	duration
xBS41	4.1	2	2	40003	9	5	1	2	4.1	Mode *load-idle signal from pressure regualtor, pB permanent*	1	yes	local	duration
xBS42	4.2	2	2	40003	10	5	2	2	4.2	Mode *load-idle signal from pressure regualtor, pA/pB change via clock*	1	yes	local	duration
xBS43	4.3	2	2	40003	11	5	3	2	4.3	Mode *load-idle signal from pressure regualtor, pA/pB change via timer*	1	yes	local	duration
xBS44	4.4	2	2	40003	12	5	4	2	4.4					
xBS45	4.5	2	2	40003	13	5	5	2	4.5	Mode *load-idle signal from pressure regualtor, pA/pB change via external contact*	1	yes	local	duration
xBS46	4.6	2	2	40003	14	5	6	2	4.6	Mode *load-idle signal from pressure regualtor, pA/pB change via SC2 master (Ethernet)*	1	yes	local	duration
xBS47	4.7	2	2	40003	15	5	7	2	4.7					

2. Master Controller ==>>>> Compressor

Table 2.1 : Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

2.x depends on model type or options, see table 3
1 always

Sigma Control 2								SAM		Description	Usability	Signal		
JSON-RPC	PROFIBUS/ PROFINET		Modbus**		Modbus	DeviceNet EtherNet/IP		Send/Receive						
BDSnd	S7 DBX	S5 DW	Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			"1"	"0"	Kind
xBS48	7.0	3	3	40004	0	6	0	3	7.0	Set speed	2.23	bus master (iDV4)	local	duration
xBS49	7.1	3	3	40004	1	6	1	3	7.1					
xBS50	7.2	3	3	40004	2	6	2	3	7.2					
xBS51	7.3	3	3	40004	3	6	3	3	7.3					
xBS52	7.4	3	3	40004	4	6	4	3	7.4					
xBS53	7.5	3	3	40004	5	6	5	3	7.5					
xBS54	7.6	3	3	40004	6	6	6	3	7.6	Enable message acknowledgement by bus master	1	yes (ack. via xBS55)	local	duration
xBS55	7.7	3	3	40004	7	6	7	3	7.7	Acknowledge signal for alarms and warnings *(if xBS54=1)*	1	*L→H:* acknowledge		flank
xBS56	6.0	3	3	40004	8	7	0	3	6.0	Enable compressor autostart by bus master	1	yes (set via xBS57)	local	duration
xBS57	6.1	3	3	40004	9	7	1	3	6.1	Compressor restart *(if xBS56=1)*	1	yes	no	duration
xBS58	6.2	3	3	40004	10	7	2	3	6.2					
xBS59	6.3	3	3	40004	11	7	3	3	6.3					
xBS60	6.4	3	3	40004	12	7	4	3	6.4					
xBS61	6.5	3	3	40004	13	7	5	3	6.5					
xBS62	6.6	3	3	40004	14	7	6	3	6.6					
xBS63	6.7	3	3	40004	15	7	7	3	6.7					

2. Master Controller ==>>>> Compressor

Table 2.1 : Receive bit assignment

2.x	depends on model type or options, see table 3
1	always

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

Sigma Control 2								SAM						
JSON-RPC	PROFIBUS/ PROFINET		Modbus**		Modbus	DeviceNet EtherNet/IP		Send/Receive		Description	Usability	Signal		
BDSnd	S7 DBX	S5 DW	Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			"1"	"0"	Kind
xBS64	35.0	17	17	40018	0	34	0	17	35.0					
xBS65	35.1	17	17	40018	1	34	1	17	35.1					
xBS66	35.2	17	17	40018	2	34	2	17	35.2					
xBS67	35.3	17	17	40018	3	34	3	17	35.3					
xBS68	35.4	17	17	40018	4	34	4	17	35.4	Set pressure pA (switching point and switching difference) via	1	bus master (iDV0+iDV1)	local	duration
xBS69	35.5	17	17	40018	5	34	5	17	35.5					
xBS70	35.6	17	17	40018	6	34	6	17	35.6	Set pressure pB (switching point and switching difference) via	1	bus master (iDV2+iDV3)	local	duration
xBS71	35.7	17	17	40018	7	34	7	17	35.7					
xBS72	34.0	17	17	40018	8	35	0	17	34.0					
xBS73	34.1	17	17	40018	9	35	1	17	34.1					
xBS74	34.2	17	17	40018	10	35	2	17	34.2					
xBS75	34.3	17	17	40018	11	35	3	17	34.3					
xBS76	34.4	17	17	40018	12	35	4	17	34.4					
xBS77	34.5	17	17	40018	13	35	5	17	34.5	Set actual air receiver system pressure via	1	bus master (iDV9)	local	duration
xBS78	34.6	17	17	40018	14	35	6	17	34.6					
xBS79	34.7	17	17	40018	15	35	7	17	34.7					

2. Master Controller ==>>>> Compressor

Table 2.1 : Receive bit assignment

2.x	depends on model type or options, see table 3
1	always

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 1 and 2

Sigma Control 2								SAM		Description	Usability	Signal		
JSON-RPC	PROFIBUS/ PROFINET		Modbus**		Modbus	DeviceNet EtherNet/IP		Send/Receive						
BDSnd	S7 DBX	S5 DW	Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			"1"	"0"	Kind
xBS80	37.0	18	18	40019	0	36	0	18	37.0					
xBS81	37.1	18	18	40019	1	36	1	18	37.1					
xBS82	37.2	18	18	40019	2	36	2	18	37.2					
xBS83	37.3	18	18	40019	3	36	3	18	37.3					
xBS84	37.4	18	18	40019	4	36	4	18	37.4					
xBS85	37.5	18	18	40019	5	36	5	18	37.5					
xBS86	37.6	18	18	40019	6	36	6	18	37.6					
xBS87	37.7	18	18	40019	7	36	7	18	37.7					
xBS88	36.0	18	18	40019	8	37	0	18	36.0					
xBS89	36.1	18	18	40019	9	37	1	18	36.1					
xBS90	36.2	18	18	40019	10	37	2	18	36.2					
xBS91	36.3	18	18	40019	11	37	3	18	36.3					
xBS92	36.4	18	18	40019	12	37	4	18	36.4					
xBS93	36.5	18	18	40019	13	37	5	18	36.5					
xBS94	36.6	18	18	40019	14	37	6	18	36.6					
xBS95	36.7	18	18	40019	15	37	7	18	36.7					

Table 3: Availability of signals for Standard machines

This table covers standard machines. In case of doubt see wiring diagram of the machine.

Avalability	SX	SM	SK	ASK	ASD	BSD	CSD	CSDX	DSD	DSDX	ESD	FSD	HSD	ASV	BSV	CSV	Description
1	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	see table 1... and 2...
2.1	x	x	x	x	●	●	●	●	●	●	●	●	●	x	●	●	Internal pressure pi
2.2	x	x	x	x	ASD.3	BSD.3	CSD.4	●	DSD.3	DSDX.3	●	FSD.3	HSD.3	x	x	x	Inlet temperature
2.3	x	x	x	x	x	x	x	1)	DSD.3	DSDX.3	●	FSD.3	HSD.3	x	x	x	Pressure discharge temperature PD
2.4	x	x	x	x	x	x	x	x	●	●	●	●	●	x	x	x	Oil separator air temperature
2.5	x	x	x	x	ASD.3	BSD.3	●	●	●	●	●	●	●	x	x	x	Motor temperature PT100
2.6	x	x	x	x	x	x	x	x	●	●	●	●	●	x	x	x	Second IOM
2.8	x	x	x	x	ASD.3	BSD.3	●	●	●	●	●	●	●	x	x	x	Motor temperature (PT100) or PTC
2.9	x	x	x	ASK.3	●	●	●	●	DSD.2A	DSDX.2A	x	FSD.2A	x	x	x	x	Oil-/air cooler fan
2.10	x	x	x	x	x	x	x	x	DSD.3A	DSDX.3A	A	FSD.3A	x	x	x	x	Oil cooler fan
2.11	x	x	x	x	x	x	x	x	DSD.3A	DSDX.3A	A	FSD.3A	x	x	x	x	Air cooler fan
2.12	x	x	x	x	x	x	x	x	W	W	W	W	●	x	x	x	Interior fan
2.13	x	x	x	x	x	x	x	x	SFC	SFC	SFC	SFC	SFC	x	x	x	Cabinet fan
2.14	x	x	x	x	x	x	x	x	x	x	x	x	●	x	x	x	HSD only
2.15	x	x	x	x	ASD.3	BSD.3	●	●	T	x	●	●	●	x	x	x	Condensate drain 1
2.16	x	x	x	x	x	x	T	T	T	x	x	x	x	x	x	x	Inlet pressure switch dryer
2.17	x	x	x	x	ASD.3T	x	x	x	T	x	x	x	x	x	x	x	Compressor motor temperature dryer
2.18	x	x	x	x	O-C2	●	●	●	●	●	●	●	●	x	●	●	Oilfilter 1 dp
2.19	x	x	x	x	O-C2	●	●	●	●	●	●	●	●	x	x	x	Air filter dp
2.20	x	x	x	x	●	●	●	●	A	A	A	A	x	x	x	x	Fan bearings
2.21	●	●	●	x	x	x	x	x	x	x	x	x	x	●	●	●	Door interlock switch
2.22	x	x	x	x	x	BSD.3	●	●	A	A	A	A	x	x	x	x	Fan bearing lube

<u>Table 3:</u> Availability of signals for Standard machines

This table covers standard machines. In case of doubt see wiring diagram of the machine.

Avalability	SX	SM	SK	ASK	ASD	BSD	CSD	CSDX	DSD	DSDX	ESD	FSD	HSD	ASV	BSV	CSV	Description
2.23	x	x	SFC with power element "SINAMICS STO "(= without mains contactor)											x	x	x	SFC machines with STO only
2.24	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	Allocatable with p,T,I at menu 5.7.2.4.x
	x	x	x	x	x	x	x	x	●	●	●	●	●	x	x	x	Allocatable with AOI2.00 (0-200 x 0,1mA)
2.25	x	x	x	x	x	x	x	x	x	x	x	FSD.3	x	x	x	x	Oilfilter 2 dp
2.26	x	x	x	x	x	x	x	x	● 2)	● 2)	● 2)	● 2)	● 2)	x	x	x	Pre-air filter dp 2)
2.C	C-machine, see wiring diagram of the machine																C-machine
2.C3	depends on machine (see wiring diagram) or on use of the function at SC 2																
2.F	S**F**C (Frequency Converter)																SFC machines only
2.O	Option, see wiring diagram of the machine																Option
2.O V	Option, see wiring diagram of the machine																Option, vacuum machines only
2.T	**T** (Dryer)																Refrigeration dryer only
2.V	**V**acuum																Vacuum machines only
2.V1	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	CSV.1	Oil pressure too high
3	depends on use of the function at SC 2																

x = not available

● = avaliable

E.g. ASD.3 = ASD 3rd generation and next

SFC = SFC machine

T = machine with dryer

1) CSDX.4 without dryer (T)

2) Software v 4.1.1 and next

A= air cooled machine

W = water cooled machine

E.g. O-C2 = machine with option C2